



CLASSROOM BOOKING SYSTEM

A seniority-aware room reservation platform for university faculty

CSE299 · Section 1(Group 2)

Supervisor: Dr. Shafin Rahman

Tawhid Hasan(2413054042) · Mohammad Tasvirul Hasan Riyad(2413692042) · Md. Tayasshuk Imam(2411600042)

INTRODUCTION

The Classroom Booking System is a full-stack reservation platform that lets faculty search, request, and manage classroom bookings online — replacing manual, spreadsheet- or email-based room scheduling.

It centers on a seniority-aware priority allocation engine: when two faculty members request the same room and time, the system resolves the conflict automatically using rank, then first-come-first-served, with waitlisting and one-click rebooking for anyone displaced.

Across 12 functional modules and 103 individual features, it covers everything from room inventory and academic calendar sync to notifications, public timetables, and analytics dashboards.

12

FUNCTIONAL MODULES

103

TOTAL FEATURES

83

MUST-HAVE REQUIREMENTS



The Problem With Manual Booking



No real-time visibility

Faculty can't see which rooms are actually free right now.



Manual, paper/email requests

Booking goes through slow back-and-forth emails or physical forms.



Frequent double-bookings

Two classes show up for the same room with no conflict check.



No fairness rules

Seniority and first-come order aren't enforced consistently.



Our Solution



Real-time availability

Search rooms instantly by date, time, capacity, equipment, and building.



Fair, rule-based allocation

Seniority-based priority with FCFS tie-breaking and conflict detection.



Automated notifications

Confirmations, waitlist promotions, and reminders sent without manual follow-up.



Data-driven oversight

Utilization, cancellation, and department reports for admins.

OBJECTIVES

- 1 Provide a single, real-time source of truth for classroom availability across the university.
- 2 Automate priority-based conflict resolution using faculty seniority, with a transparent, fair fallback rule.
- 3 Support one-time and recurring bookings with full lifecycle management — request, confirm, edit, cancel, rebook.
- 4 Keep every stakeholder informed through automated notifications and reminders.
- 5 Give administrators visibility through utilization dashboards and exportable reports.
- 6 Expose a public, read-only timetable so students and staff can check room schedules without an account.

TECH STACK



Frontend

- Next.js (React)
- TypeScript
- Responsive, component-based UI
- Public timetable & dashboard views



Backend

- Node.js + Express
- TypeScript throughout
- REST API layer
- Session & role-based auth



Database

- PostgreSQL
- Schema migrations
- Relational modeling for bookings, rooms, faculty ranks
- Audit-friendly booking history



System at a Glance

12 modules · 103 functional features — a closer look at each one follows



Classroom Inventory

7 features



Faculty & Seniority

6 features



Availability & Search

8 features



Booking Request

10 features



Priority Allocation

12 features



Recurring Booking

8 features



Academic Calendar

7 features



Notification

10 features



Booking Management

9 features



Timetable & Public View

8 features



Reporting & Analytics

8 features

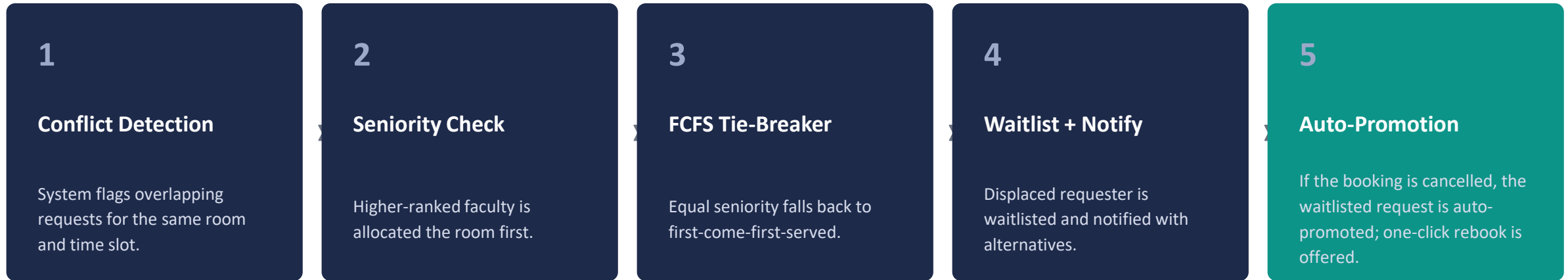


Admin Configuration

10 features

DEEP DIVE: PRIORITY ALLOCATION

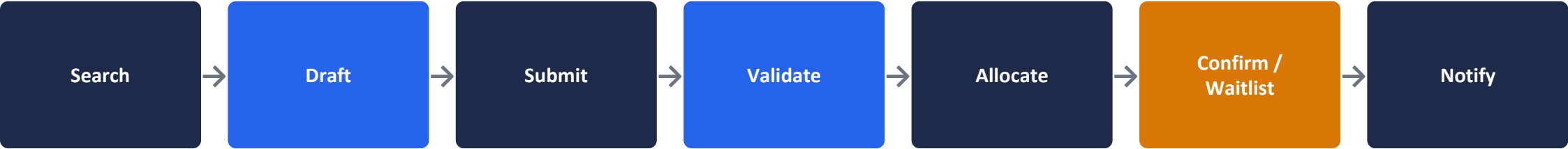
The core logic that resolves booking conflicts fairly and automatically.



Related safeguards

- No-Bump policy — a confirmed booking can't be displaced once locked in
- Department Head override — for exceptional cases, with a logged override notification
- Booking window enforcement — requests are only accepted within an allowed lead time

BOOKING LIFECYCLE: REQUEST TO CONFIRMATION



Booking validation checks capacity, equipment requirements, and duplicate/overlap conditions before a request ever reaches priority allocation. Every stage writes to booking history for auditability.

Booking ID generation

Every confirmed booking gets a unique, trackable ID used across notifications, check-in, and reports.

QR check-in

Faculty scan a QR code to confirm attendance, feeding directly into no-show detection.

No-show handling

Unclaimed rooms are flagged and can be auto-promoted to the next waitlisted request.



Admin Configuration

- System settings
- Booking rules
- Operating hours
- Room type rules
- Role management
- Notification settings
- Calendar settings
- Booking window configuration
- User management
- System configuration controls



Demo Walkthrough



Search

Faculty searches rooms by date,
time & capacity



Book

Submits a one-time or recurring
booking request



Allocate

System validates, checks conflicts
& applies priority



Notify

Confirmation, waitlist or
reminder is sent automatically



Check-in

QR check-in on arrival; usage
logged for reports



Live Application Walkthrough

Real screens from the running system — no mockups



Secure Institutional Login

Screen 1 of 5 · Role-based SSO access

1/5

- Unified sign-in screen for every role: Registrar Admin, Department Head, Faculty and read-only Student access
- Demo bypass profiles switch role context instantly for grading and testing, no password required
- Live API health check and server timestamp shown for transparency

The screenshot shows a dark-themed login interface for the 'Classroom Booking System'. At the top, there's a toggle for 'Institutional SSO Portal' and a 'Light' theme switch. The title 'Classroom Booking System' is prominent, followed by a subtitle: 'Provide credentials or choose a bypass profile to access resources.' Below this, the 'SECURE LOGIN' section contains input fields for 'EMAIL / USERNAME' (with a hint 'e.g. admin or faculty email') and 'PASSWORD' (masked with dots). A bright blue 'Sign In' button is positioned below these fields. Underneath, the 'DEMO SSO BYPASS PROFILES' section features a dropdown menu labeled '-- Choose Profile for Instant Login --'. The dropdown is open, showing a list of roles: 'Registrar (System Admin)', 'Dr. Arshad (ECE Dept Head)', 'Dr. Nusrat (ECE Faculty)', 'Samiul Hasan (Lecturer)', and 'Read-only Student Portal'. At the bottom, there are two informational boxes: 'API' showing the URL 'https://cse-299-classroom-booking-system.onrender.com/api/health' and 'SERVER TIME' showing '2026-07-25T17:29:27.217Z'.

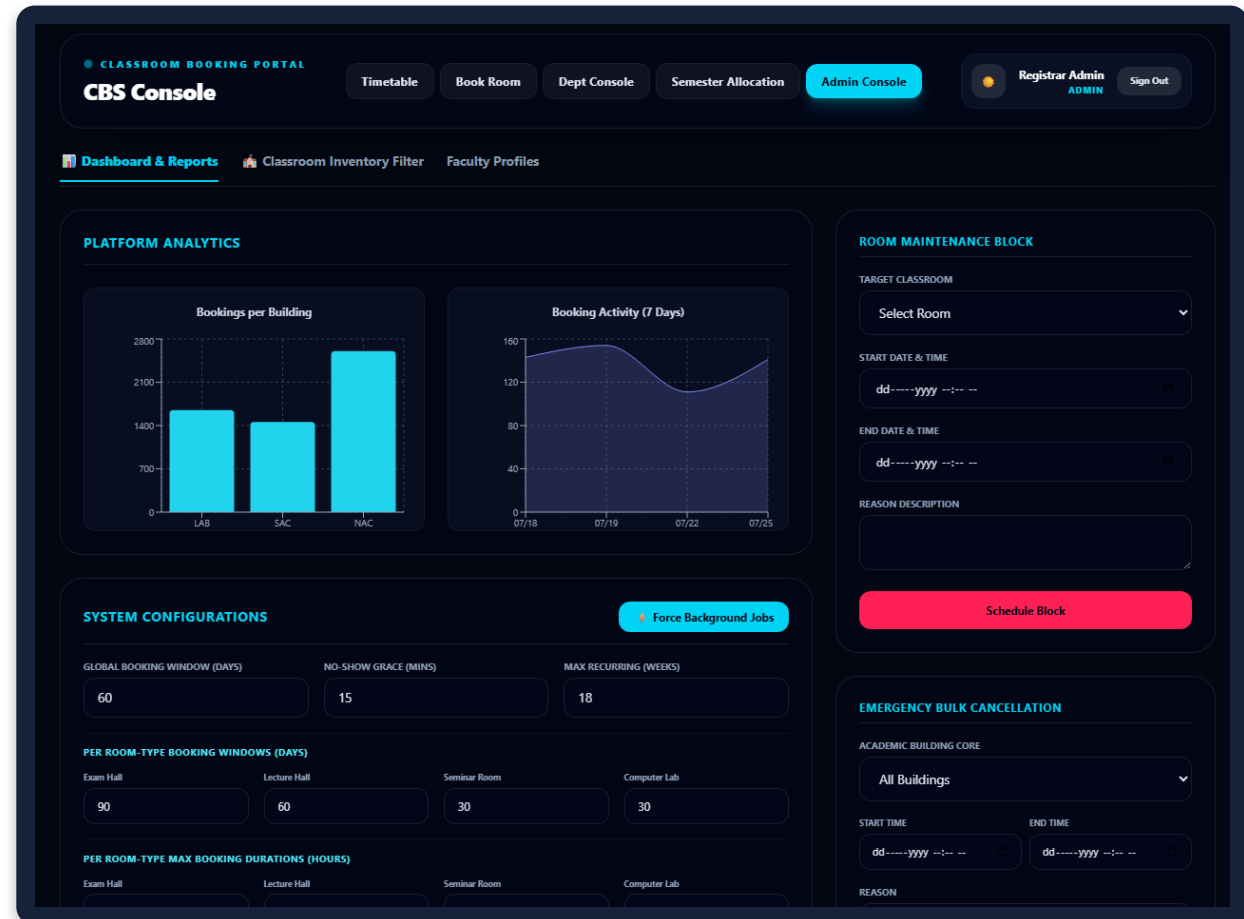


Admin Dashboard & Platform Analytics

2/5

Screen 2 of 5 · Registrar Admin Console

- Live charts for bookings-per-building and 7-day booking activity
- Global and per room-type booking windows, no-show grace period and max recurring weeks, all admin-configurable
- Room maintenance blocking and emergency bulk cancellation tools for the whole academic building core





Bulk Semester Allocation Engine

Screen 3 of 5 · Priority-based batch scheduling

3/5

- Spring 2026 batch shown mid-run: 72 of 204 sections allocated, 132 conflicts flagged for review
- Each row shows the allocation basis, e.g. Department Mandate or Senior Preference Fallback
- One click to publish the draft to the public timetable or confirm & create live bookings

CLASSROOM BOOKING PORTAL

CBS Console

TimetableBook RoomDept ConsoleSemester AllocationAdmin Console

Registrar AdminADMINSign Out

Bulk Semester Allocation Console

Draft, optimize, resolve scheduling conflicts, and finalize the university semester timetable.

← Back to Batches Directory

SEMESTER

Spring2026

ID: 9c4e472b-a278-4294-a18f-8552ab43022

ALLOCATED SUCCESS

72 / 204

Sections mapped to classrooms

ACTIVE CONFLICTS

132

Sections requiring review

BATCH STATUS

REVIEWCONFLICTS

Allocation Engine Coverage

35%

Allocated List (72)

Conflicts List (132)

Faculty PDF

Room PDF

Full Report

Publish to Timetable

Export Draft CSV

Confirm & Create Bookings

COURSE	SECTION	SIZE	ROOM ASSIGNED	FACULTY ASSIGNED	TIME SLOT	ALLOCATION BASIS	ACTIONS
CSE201 Data Structures	01	SAC-201 (40 seats)	Dr. Nusrat Jahan AssociateProfessor	MW (Mon & Wed) 11:30 - 13:00	Department Mandate: Dr. Nusrat Jahan	ClassFacultyAdjust	
MAT116 Precalculus	01	NAC-101 (50 seats)	Dr. Kazi Mahbub Professor	MW (Mon & Wed) 19:40 - 21:10	Senior Preference Fallback	ClassFacultyAdjust	
MAT120 Calculus and Analytic Geometry I	01	SAC-101 (50 seats)	Dr. Kazi Mahbub Professor	MW (Mon & Wed) 19:40 - 21:10	Senior Preference Fallback	ClassFacultyAdjust	
CSE311 Database Systems	01	NAC-402 (40 seats)	Dr. Arshad Rahman Professor	ST (Sun & Tue) 13:00 - 14:30	Senior Preference Fallback	ClassFacultyAdjust	
CSE225 Algorithms	01	SAC-201 (40 seats)	Dr. Arshad Rahman Professor	MW (Mon & Wed) 18:00 - 19:30	Senior Preference Fallback	ClassFacultyAdjust	



Faculty Booking Portal

Screen 4 of 5 · Faculty dashboard

4/5

- Personal booking statistics: active, waitlisted, cancelled and auto no-show counts at a glance
- Course Preference Console for ranked section requests, with last-submitted choices preserved
- Confirmed semester schedule with PDF export and automated reminder notifications

The screenshot displays the 'CBS Console' interface for a faculty member, Dr. Arshad Rahman. The top navigation bar includes links for 'Timetable', 'Book Room', and 'My Bookings'. The main content area is divided into several sections:

- Your Bookings & Requests:** A section for managing booking status, currently showing 'No reservation records found. Make a booking request to begin.'
- Course Preference Console:** A section indicating that preference submissions are currently closed. It shows a list of 'YOUR LAST SUBMITTED CHOICES' with ranked options for various courses and sections.
- MY BOOKING STATISTICS:** A dashboard showing counts for 'ACTIVE CONFIRMED', 'WAITLISTED QUEUE', 'CANCELLED RUN', and 'AUTO NO SHOW'.
- MY SEMESTER SCHEDULE:** A section displaying allocated sections for the current semester, including course details, section numbers, and times.
- NOTIFICATION LOGS:** A section showing simulated institutional email alerts and confirmations, including an 'Upcoming Booking Reminder' for Room SAC-201.



Admin Configuration & Inventory Tools

Screen 5 of 5 · Registrar Admin Console

5/5

- Opens or closes the faculty preference collection window per target semester
- Pending admin approvals and active classroom alerts surfaced in one place
- Add classroom records individually or bulk import an entire inventory via Excel/CSV

The screenshot displays the 'Admin Console' interface of the 'CLASSROOM BOOKING PORTAL CBS Console'. The top navigation bar includes links for 'Timetable', 'Book Room', 'Dept Console', 'Semester Allocation', and 'Admin Console' (which is highlighted). A user profile for 'Registrar Admin' is visible in the top right corner.

The main content area is divided into several sections:

- FACULTY PREFERENCE COLLECTION WINDOW**: A section with a 'CLOSED' status. It includes fields for 'TARGET SEMESTER' (Summer26) and 'SUBMISSION DEADLINE' (25-Jul-2026 05:43 PM). Below these is a text area for 'INSTRUCTIONS FOR FACULTY (OPTIONAL)' with the example text: 'e.g. Please submit your top 3 course preferences. Senior faculty choices will be prioritised.' There are two buttons: 'Open Preference Window' and 'Update Settings'.
- PENDING ADMIN APPROVALS QUEUE (0)**: A section showing 'No pending requests require admin approval.'
- ACTIVE CLASSROOM ALERTS & ISSUES (0)**: A section showing 'No active room issues reported today.'
- OCCUPANCY & BOOKING REPORTS**: A section with a 'ROOM UTILIZATION BREAKDOWN' heading.

On the right side of the interface, there are two additional sections:

- ADD NEW CLASSROOM RECORD**: A form with fields for 'ROOM NUMBER' (e.g. NAC-501), 'BUILDING' (Select Building), 'FLOOR' (5), 'CAPACITY' (45), and 'TYPE' (Lecture Hall). It also has checkboxes for 'EQUIPMENT' (Projector, Whiteboard, Smart Board, Air Conditioning) and an 'Add Classroom Record' button.
- INVENTORY BULK IMPORT (EXCEL / CSV)**: A section with a 'Download Excel Template' link. It includes an 'UPLOAD SPREADSHEET FILE (XLSX, XLS, .CSV)' area with a 'Choose File' button and a 'No file chosen' message. Below this is an 'OR EDIT RAW JSON PREVIEW' section showing a JSON snippet for a room record.



Feature Breakdown

103 functional features prioritized using MoSCoW

103

Total features

12

System modules

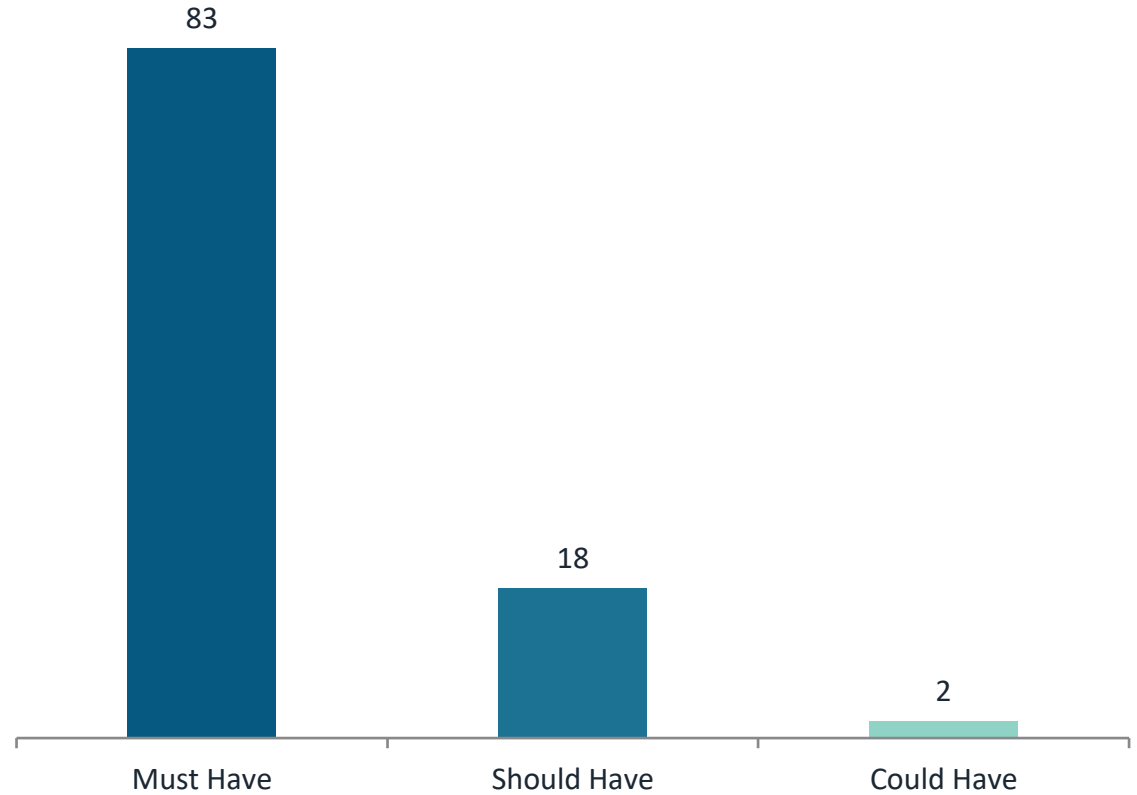
How we read this

Must Have (83): core booking, allocation & notification flow.

Should Have (18): reporting, analytics & admin conveniences.

Could Have (2): nice-to-have polish for later releases.

Features by Priority (MoSCoW)



CHALLENGES FACED



Local environment setup

Coordinating SSH authentication, PostgreSQL configuration, and consistent tooling across teammates' machines.



Schema migration conflicts

A missing base schema file had to be reverse-engineered before migrations would apply cleanly across the team.



Priority logic edge cases

Designing seniority + FCFS + override rules that stay predictable even under overlapping recurring bookings.



Full-stack integration

Keeping the Express/TypeScript API, Next.js frontend, and PostgreSQL schema in sync as features were built in parallel.

FUTURE ROADMAP



Mobile-Friendly Booking

A responsive or native mobile flow so faculty can request and check bookings on the go.



Predictive Utilization Insights

Use historical booking data to recommend optimal room assignments and flag underused spaces.



Calendar App Integration

Two-way sync with Google/Outlook calendars for confirmed bookings.



Smarter Notifications

Configurable channels (SMS/push) and personalized reminder timing.



Multi-Campus Support

Extend room inventory and booking rules across multiple campuses or buildings.

CONCLUSION

Manual, spreadsheet-driven room scheduling doesn't scale with a growing faculty and a packed academic calendar — conflicts, double-bookings, and unfair allocation are the norm rather than the exception.

The Classroom Booking System replaces that with a transparent, automated allocation process — seniority-aware, conflict-free, and fully auditable — while giving administrators the reporting tools to keep improving room utilization over time.



**12 modules, 103 features, one
consistent allocation model.**



Thank You

Classroom Booking System